PRACTICAL NO. 2

Diagnostic Analytics Charts

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Course: Data Analytics and Visualization

Course Code: CT-471

Practical Overview

This practical covers diagnostic analytics — going beyond 'what happened' to understand 'why it happened.' Using scatter charts, bubble charts, ribbon charts, waterfall charts, and funnel charts, we analyze sales efficiency, ranking dynamics, contribution patterns, and market share trends across the retail dataset (2023–2024).

Visual Type	Questions
Scatter Chart	Q1, Q2, Q3, Q16
Bubble Chart	Q4, Q5, Q6
Ribbon Chart	Q7, Q8, Q9
Waterfall Chart	Q10, Q11, Q12
Funnel Chart	Q13, Q14, Q15
Advanced Visuals	Q17, Q18, Q19, Q20

Questions 1–20

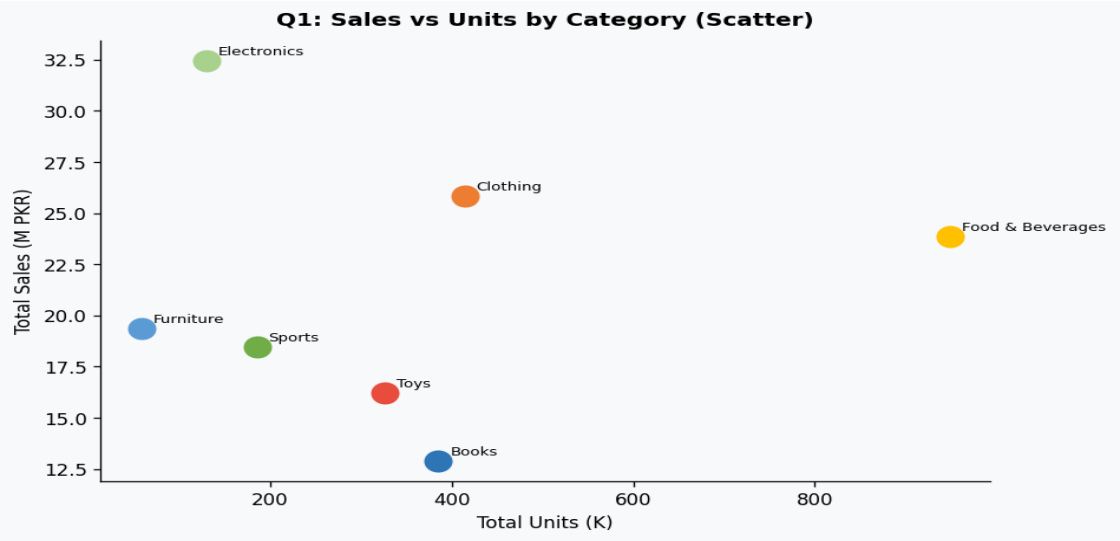
Q1 – Scatter Chart: Sales vs Units by Category

Analyze the relationship between total sales value and total units sold across product categories to identify high-revenue, high-volume leaders versus niche premium categories.

SQL Query

```
SELECT CATEGORY,
       SUM(SALES)                                  AS TOTAL_SALES,
       SUM(UNITS)                                  AS TOTAL_UNITS
FROM   RETAIL_TRANSACTIONS
GROUP  BY CATEGORY
ORDER  BY TOTAL_SALES DESC;
```

Visualization: Scatter Chart



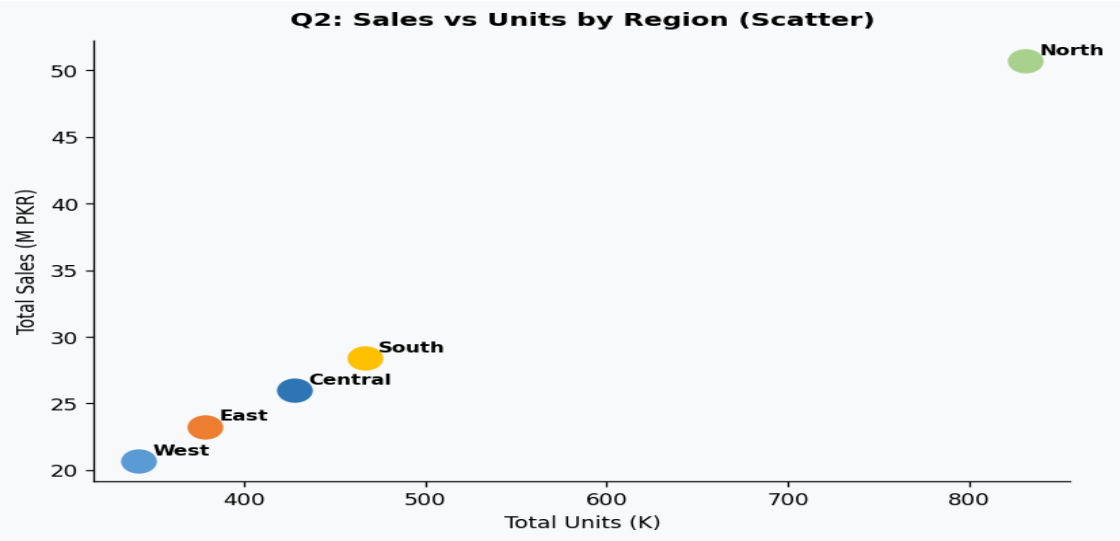
Q2 – Scatter Chart: Sales vs Units by Region

Compare regional sales value against units sold to identify regions with high revenue efficiency (high sales, moderate units) versus volume-driven regions.

SQL Query

```
SELECT REGION,
       SUM(SALES)   AS TOTAL_SALES,
       SUM(UNITS)   AS TOTAL_UNITS
FROM   RETAIL_TRANSACTIONS
GROUP  BY REGION
ORDER  BY TOTAL_SALES DESC;
```

Visualization: Scatter Chart



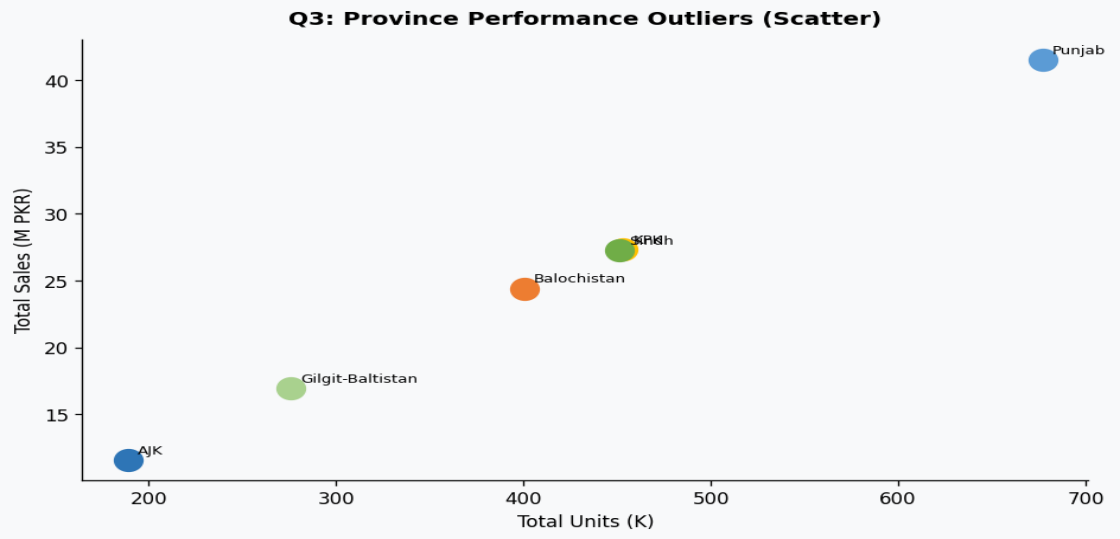
Q3 – Scatter Chart: Province-Level Performance Outliers

Plot province-level sales against units to detect outliers — provinces that over- or under-perform relative to their volume.

SQL Query

```
SELECT STATE_PROVINCE,
       SUM(SALES)   AS TOTAL_SALES,
       SUM(UNITS)   AS TOTAL_UNITS
FROM   RETAIL_TRANSACTIONS
GROUP  BY STATE_PROVINCE
ORDER  BY TOTAL_SALES DESC;
```

Visualization: Scatter Chart



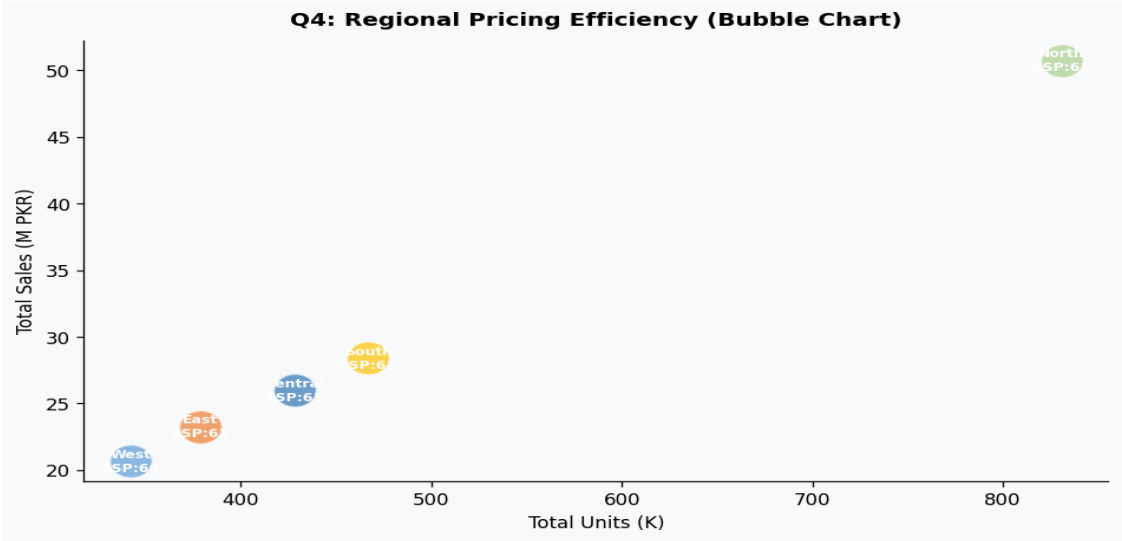
Q4 – Bubble Chart: Regional Performance with Pricing Efficiency

Extend regional analysis by adding Average Selling Price as the bubble size, revealing the three-dimensional relationship between revenue, volume, and pricing.

SQL Query

```
SELECT REGION,
       SUM(SALES)                                        AS TOTAL_SALES,
       SUM(UNITS)                                        AS TOTAL_UNITS,
       ROUND(SUM(SALES) * 100.0 / SUM(SUM(SALES)) OVER (), 2) AS AVG_SELLING_PRICE
FROM   RETAIL_TRANSACTIONS
GROUP  BY REGION
ORDER  BY TOTAL_SALES DESC;
```

Visualization: Bubble Chart



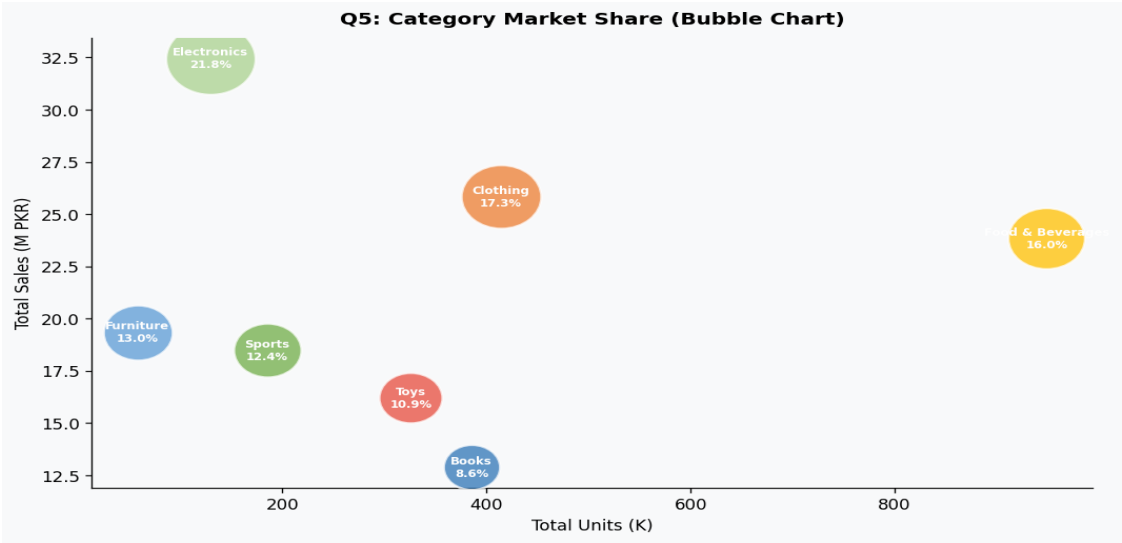
Q5 – Bubble Chart: Category Performance and Market Share

Bubble chart showing each category's sales, units, and market share percentage (bubble size) for a complete dominance picture.

SQL Query

```
SELECT CATEGORY,
       SUM(SALES)                                          AS TOTAL_SALES,
       SUM(UNITS)                                         AS TOTAL_UNITS,
       ROUND(SUM(SALES)*100.0 / SUM(SUM(SALES)) OVER (), 2)  AS MARKET_SHARE_PCT
FROM   RETAIL_TRANSACTIONS
GROUP  BY CATEGORY
ORDER  BY TOTAL_SALES DESC;
```

Visualization: Bubble Chart



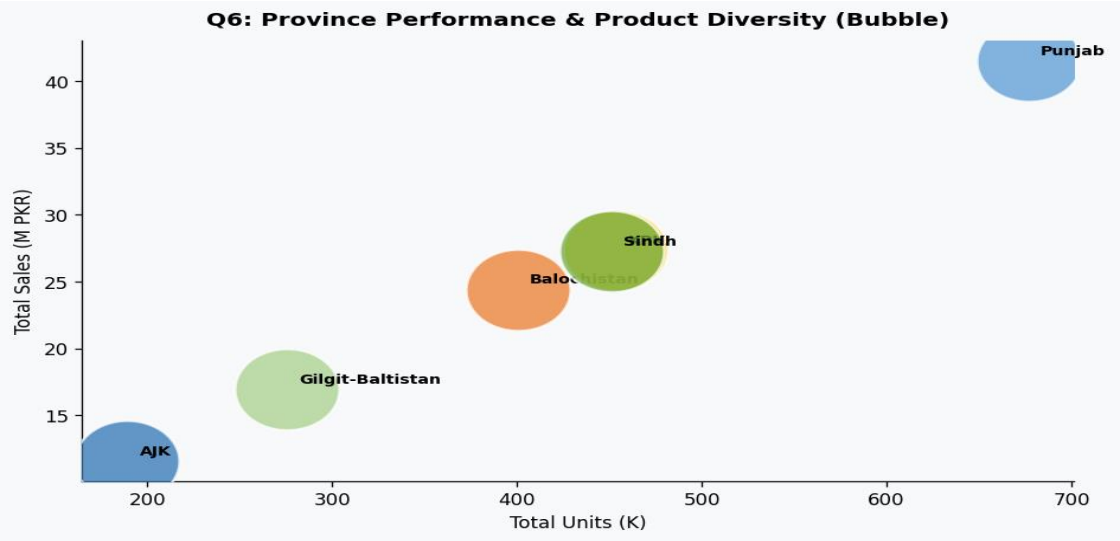
Q6 – Bubble Chart: Province Performance with Product Diversity

Province-level bubble showing total sales, units, and product diversity (distinct categories sold) as the bubble size.

SQL Query

```
SELECT STATE_PROVINCE,
       SUM(SALES)                                        AS TOTAL_SALES,
       SUM(UNITS)                                       AS TOTAL_UNITS,
       COUNT(DISTINCT CATEGORY)                         AS CATEGORY_COUNT
FROM   RETAIL_TRANSACTIONS
GROUP  BY STATE_PROVINCE
ORDER  BY TOTAL_SALES DESC;
```

Visualization: Bubble Chart



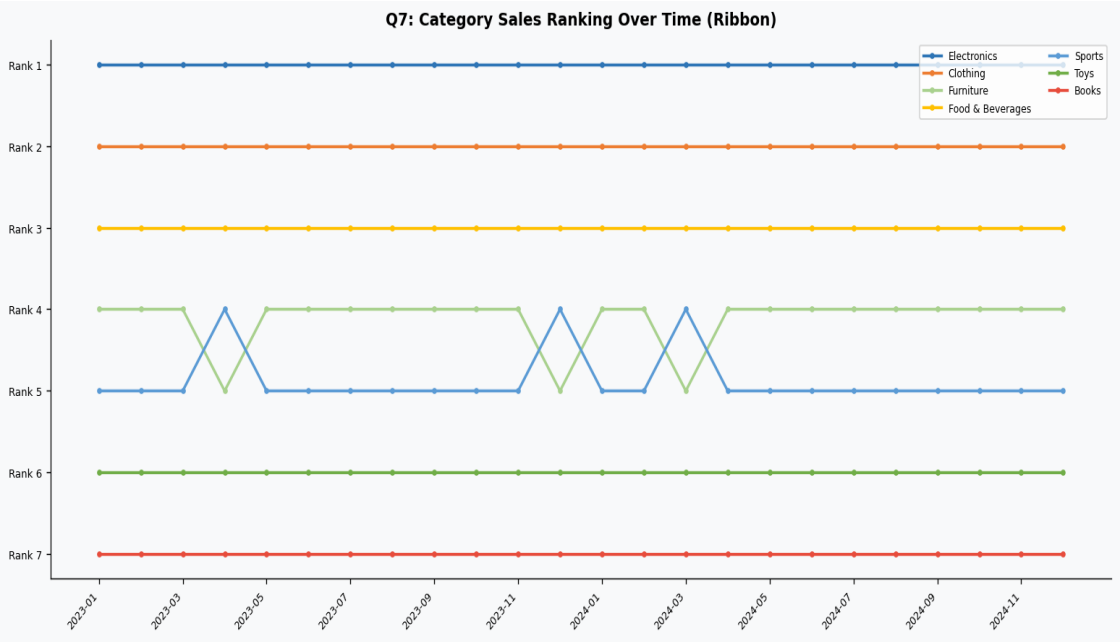
Q7 – Ribbon Chart (Rank Over Time): Category Ranking Change Over Time

Track how category sales rankings shift month-to-month across 2023–2024 using a ribbon-style rank line chart.

SQL Query

```
SELECT YEAR, MONTH, CATEGORY,
       SUM(SALES)                                        AS MONTHLY_SALES,
       RANK() OVER (PARTITION BY YEAR, MONTH
                    ORDER BY SUM(SALES) DESC)            AS SALES_RANK
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, MONTH, CATEGORY
ORDER  BY YEAR, MONTH, SALES_RANK;
```

Visualization: Ribbon Chart (Rank Over Time)



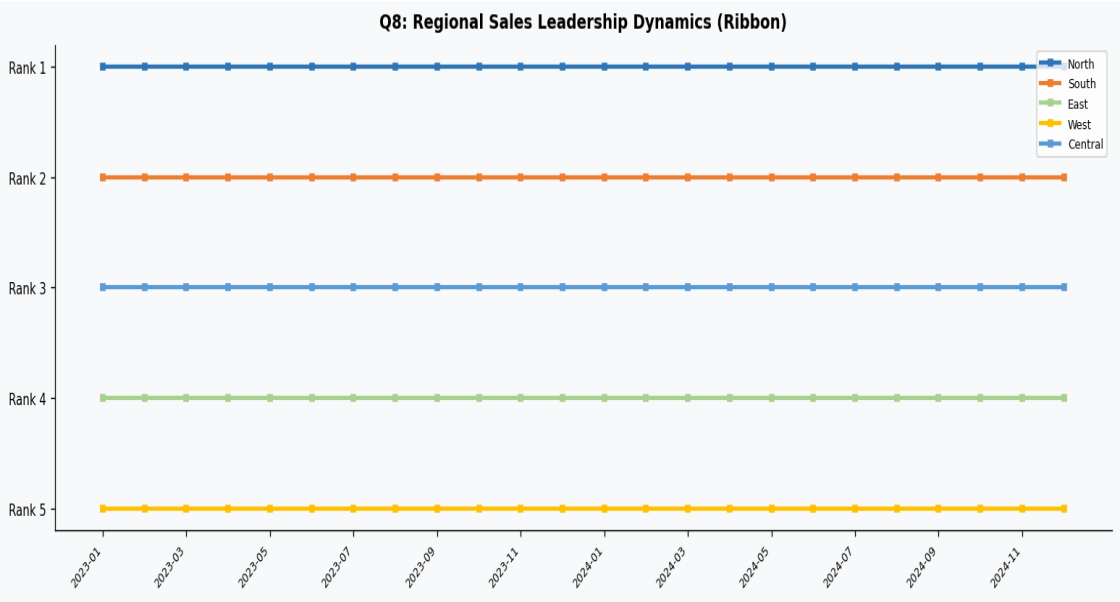
Q8 – Ribbon Chart (Rank Over Time): Regional Leadership Dynamics

Monitor regional leadership changes over time. Identify which region consistently leads and where rank reversals occur.

SQL Query

```
SELECT YEAR, MONTH, REGION,
       SUM(SALES)                                          AS MONTHLY_SALES,
       RANK() OVER (PARTITION BY YEAR, MONTH
                    ORDER BY SUM(SALES) DESC)              AS SALES_RANK
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, MONTH, REGION
ORDER  BY YEAR, MONTH, SALES_RANK;
```

Visualization: Ribbon Chart (Rank Over Time)



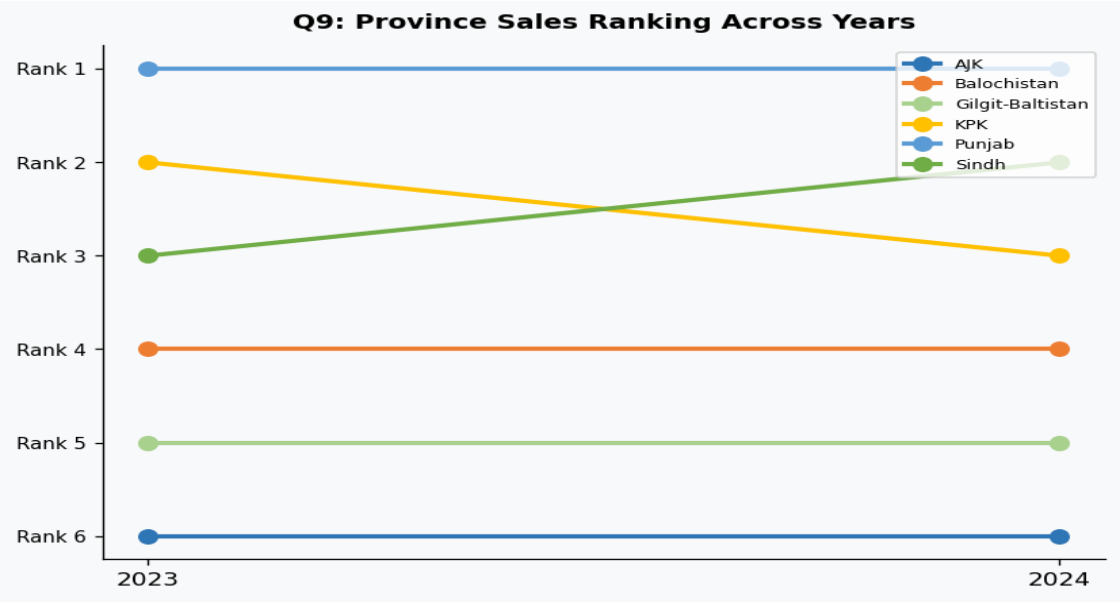
Q9 – Ribbon Chart (Yearly Rank Shift): Province Ranking Across Years

Compare each province's annual ranking between 2023 and 2024 to reveal long-term competitive shifts.

SQL Query

```
SELECT YEAR, STATE_PROVINCE,
       SUM(SALES)                                          AS YEARLY_SALES,
       RANK() OVER (PARTITION BY YEAR
                    ORDER BY SUM(SALES) DESC)              AS SALES_RANK
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, STATE_PROVINCE
ORDER  BY YEAR, SALES_RANK;
```

Visualization: Ribbon Chart (Yearly Rank Shift)



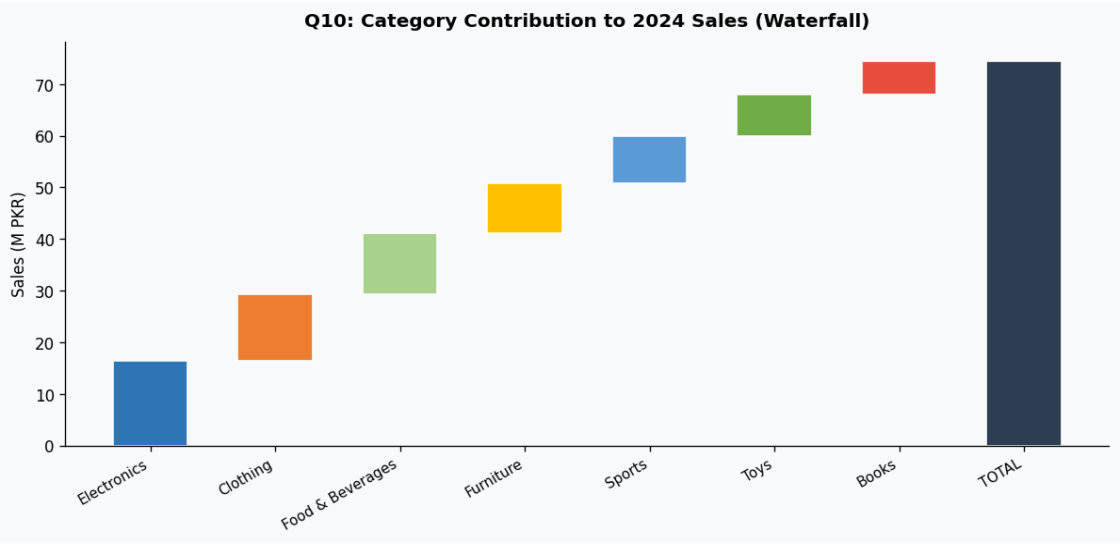
Q10 – Waterfall Chart: Category Contribution to Yearly Sales

Waterfall chart showing how each category's sales stack up to form annual total. Each bar represents an additive contribution.

SQL Query

```
SELECT YEAR, CATEGORY,
       SUM(SALES)                                          AS CATEGORY_SALES,
       SUM(SUM(SALES)) OVER (PARTITION BY YEAR)           AS YEARLY_TOTAL
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, CATEGORY
ORDER  BY YEAR, CATEGORY_SALES DESC;
```

Visualization: Waterfall Chart



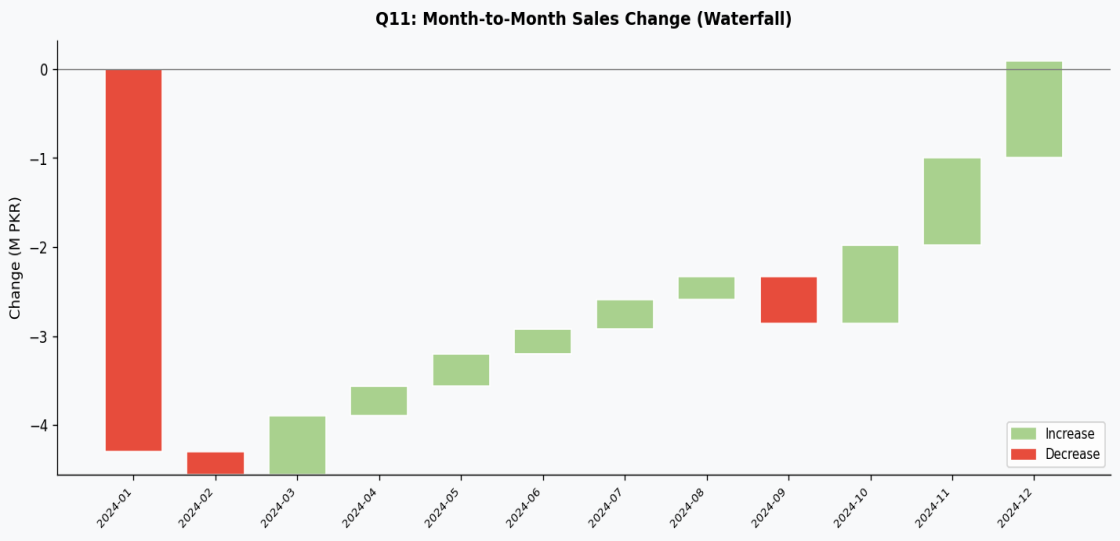
Q11 – Waterfall Chart: Month-to-Month Sales Change

Display month-over-month changes in total sales. Green bars represent growth months; red bars represent decline months.

SQL Query

```
SELECT YEAR, MONTH,
       SUM(SALES)                                          AS MONTHLY_SALES,
       SUM(SALES) - LAG(SUM(SALES))
                      OVER (ORDER BY YEAR, MONTH)          AS MOM_CHANGE
FROM   RETAIL_TRANSACTIONS
GROUP  BY YEAR, MONTH
ORDER  BY YEAR, MONTH;
```

Visualization: Waterfall Chart



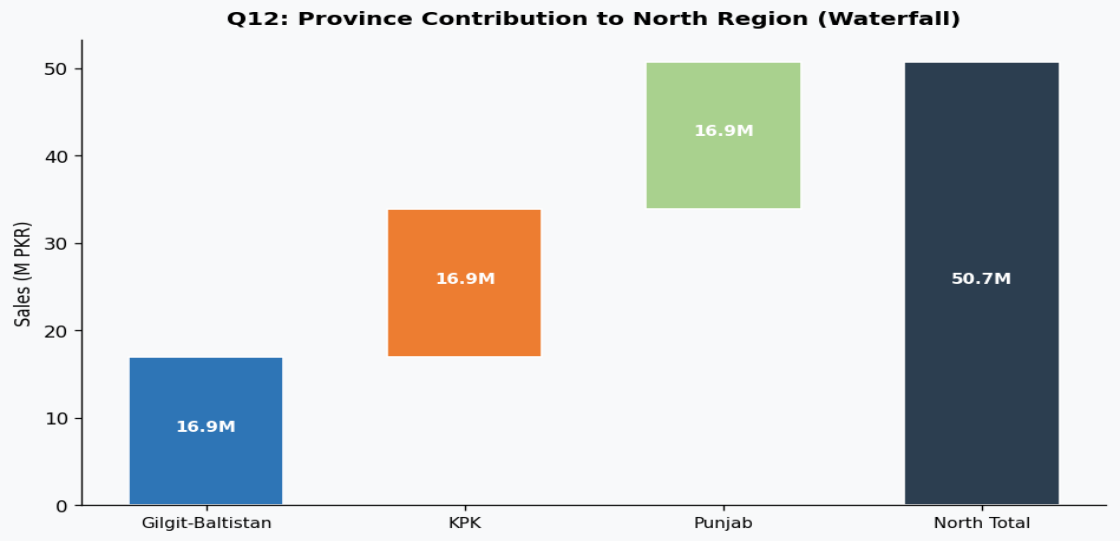
Q12 – Waterfall Chart: Provincial Contribution to Regional Sales

Show how each province within the North region contributes cumulatively to the regional total using a waterfall build-up chart.

SQL Query

```
SELECT REGION, STATE_PROVINCE,
       SUM(SALES)                                          AS PROVINCE_SALES,
       SUM(SUM(SALES)) OVER (PARTITION BY REGION)          AS REGIONAL_TOTAL
FROM   RETAIL_TRANSACTIONS
GROUP  BY REGION, STATE_PROVINCE
ORDER  BY REGION, PROVINCE_SALES DESC;
```

Visualization: Waterfall Chart



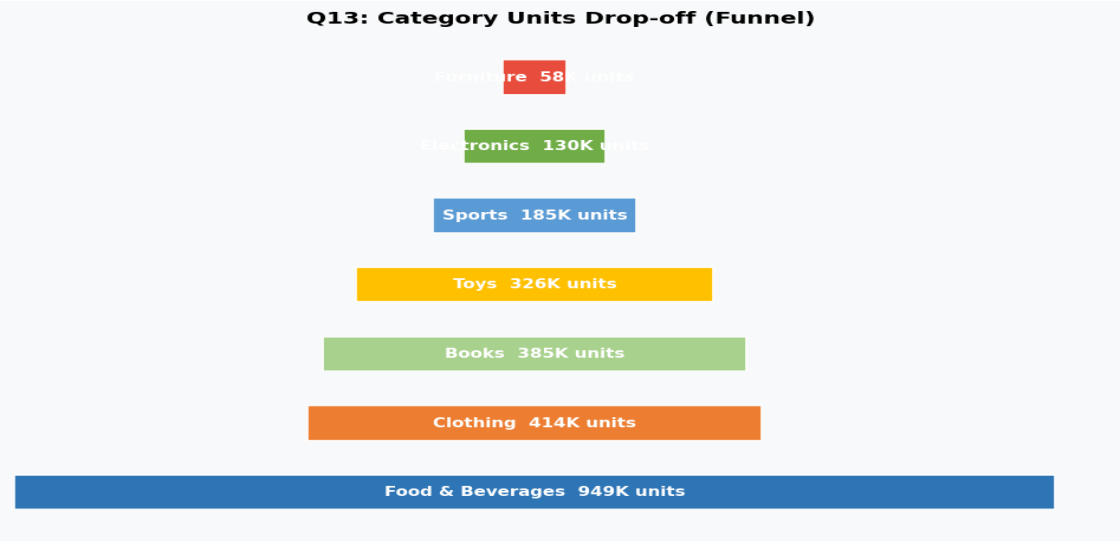
Q13 – Funnel Chart: Category-Based Sales Drop-Off

Funnel chart ordered by category units — illustrates the 'drop-off' in volume from the highest-selling category downward.

SQL Query

```
SELECT CATEGORY,
       SUM(UNITS)                                          AS TOTAL_UNITS,
       ROUND(100.0 * (1 - SUM(UNITS)*1.0 /
             FIRST_VALUE(SUM(UNITS))
               OVER (ORDER BY SUM(UNITS) DESC)), 2)        AS PCT_DROP
FROM   RETAIL_TRANSACTIONS
GROUP  BY CATEGORY
ORDER  BY TOTAL_UNITS DESC;
```

Visualization: Funnel Chart



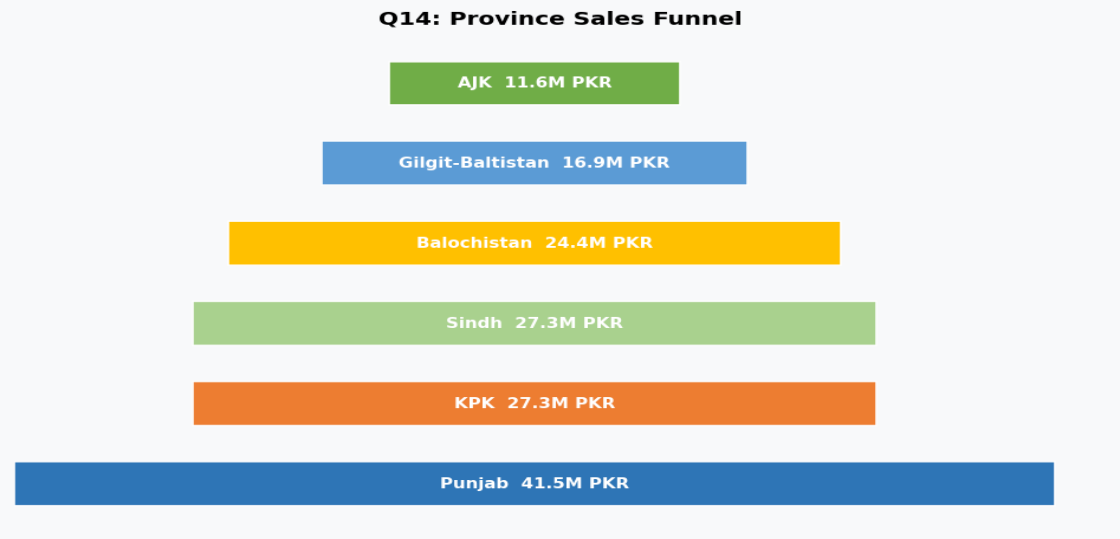
Q14 – Funnel Chart: Province-Based Sales Reduction

Province-level sales funnel showing the tapering of revenue from top to bottom province.

SQL Query

```
SELECT STATE_PROVINCE,
       SUM(SALES)                                          AS TOTAL_SALES,
       ROUND(100.0 * (1 - SUM(SALES)*1.0 /
             FIRST_VALUE(SUM(SALES))
               OVER (ORDER BY SUM(SALES) DESC)), 2)        AS PCT_DROP
FROM   RETAIL_TRANSACTIONS
GROUP  BY STATE_PROVINCE
ORDER  BY TOTAL_SALES DESC;
```

Visualization: Funnel Chart



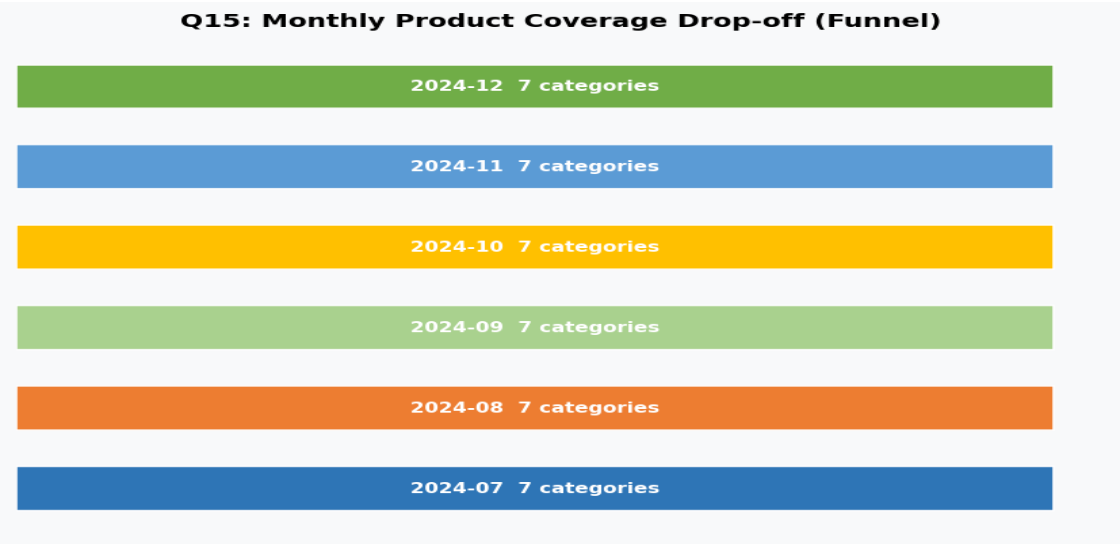
Q15 – Funnel Chart: Time-Based Product Coverage Drop-Off

Track the number of distinct categories sold each month and show the percentage drop relative to the previous month.

SQL Query

```
WITH MonthCoverage AS (
  SELECT YEAR, MONTH,
         COUNT(DISTINCT CATEGORY)                              AS CAT_COUNT
  FROM   RETAIL_TRANSACTIONS
  GROUP  BY YEAR, MONTH
)
SELECT YEAR, MONTH, CAT_COUNT,
       ROUND(100.0 * (1 - CAT_COUNT*1.0 /
             LAG(CAT_COUNT) OVER (ORDER BY YEAR,MONTH)), 2)   AS PCT_DROP
FROM   MonthCoverage
ORDER  BY YEAR, MONTH;
```

Visualization: Funnel Chart



Q16 – Scatter Chart (Efficiency vs Growth): Category Efficiency vs Growth Trade-off

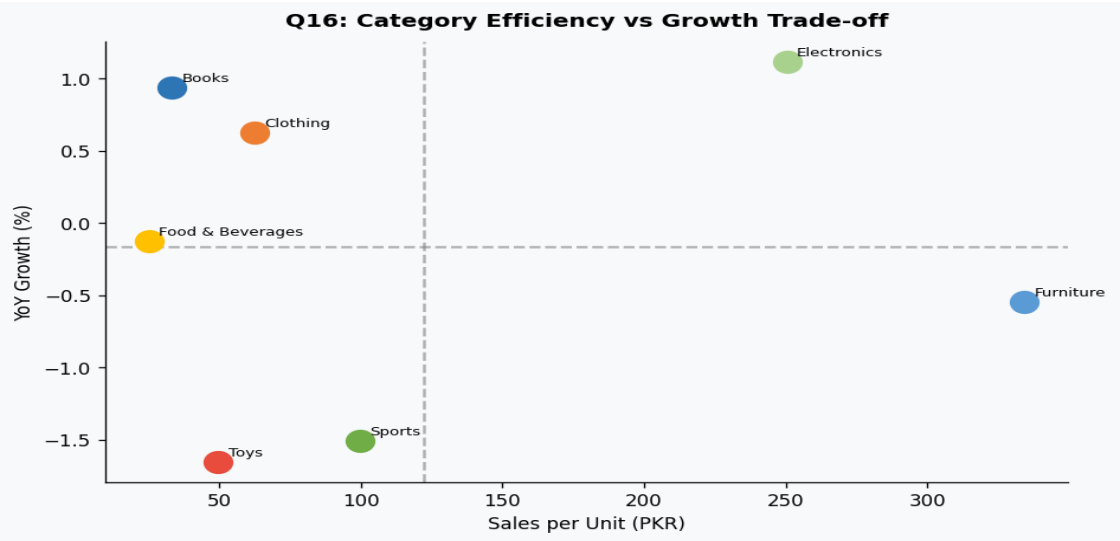
Scatter chart plotting each category's sales per unit (efficiency) on the x-axis against year-over-year growth rate on y-axis.

SQL Query

```
WITH CategoryMetrics AS (
  SELECT CATEGORY,
         SUM(SALES) / NULLIF(SUM(UNITS),0)                     AS SALES_PER_UNIT,
```

```
        SUM(UNITS)                                          AS TOTAL_UNITS
  FROM   RETAIL_TRANSACTIONS
  GROUP  BY CATEGORY
),
GrowthCalc AS (
  SELECT CATEGORY, YEAR, SUM(SALES) AS YR_SALES
  FROM   RETAIL_TRANSACTIONS
  GROUP  BY CATEGORY, YEAR
)
SELECT c.CATEGORY, c.SALES_PER_UNIT, c.TOTAL_UNITS,
       ROUND((g24.YR_SALES - g23.YR_SALES)*100.0 /
             NULLIF(g23.YR_SALES,0), 2)                     AS GROWTH_RATE
FROM   CategoryMetrics c
JOIN   GrowthCalc g23 ON c.CATEGORY=g23.CATEGORY AND g23.YEAR=2023
JOIN   GrowthCalc g24 ON c.CATEGORY=g24.CATEGORY AND g24.YEAR=2024;
```

Visualization: Scatter Chart (Efficiency vs Growth)



Q17 – Bar + Error Bars (Volatility): Regional Sales Volatility Detection

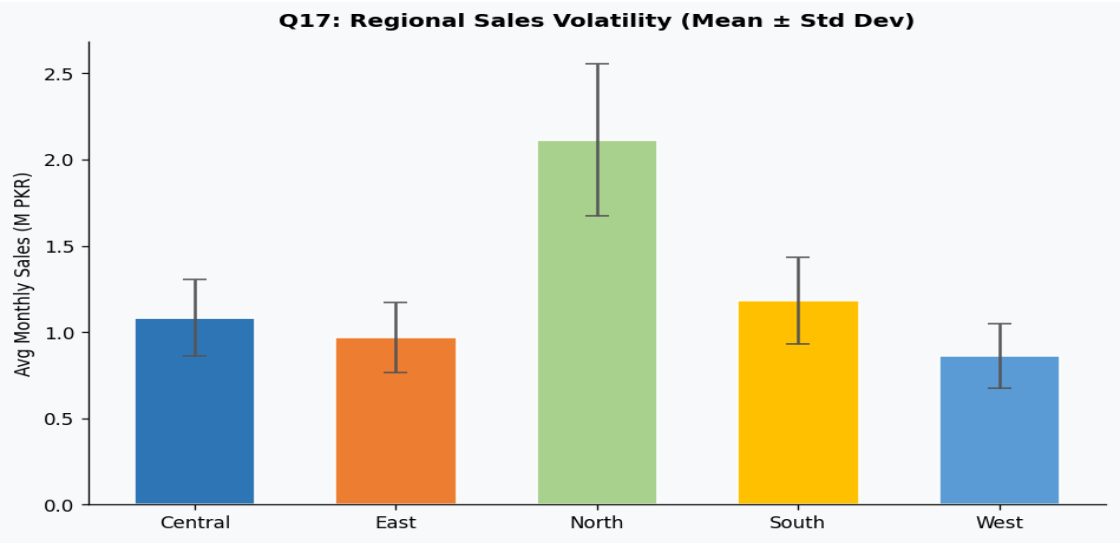
Bar chart with error bars (±1 Std Dev) per region, showing which regions have stable monthly sales versus highly volatile ones.

SQL Query

```
SELECT REGION,
       STDDEV(MONTHLY_SALES)                                AS SALES_STD_DEV,
       AVG(MONTHLY_SALES)                                   AS AVG_MONTHLY_SALES
FROM (
  SELECT REGION, YEAR, MONTH, SUM(SALES) AS MONTHLY_SALES
  FROM   RETAIL_TRANSACTIONS
  GROUP  BY REGION, YEAR, MONTH
) t
```

```
GROUP  BY REGION
ORDER  BY SALES_STD_DEV DESC;
```

Visualization: Bar + Error Bars (Volatility)



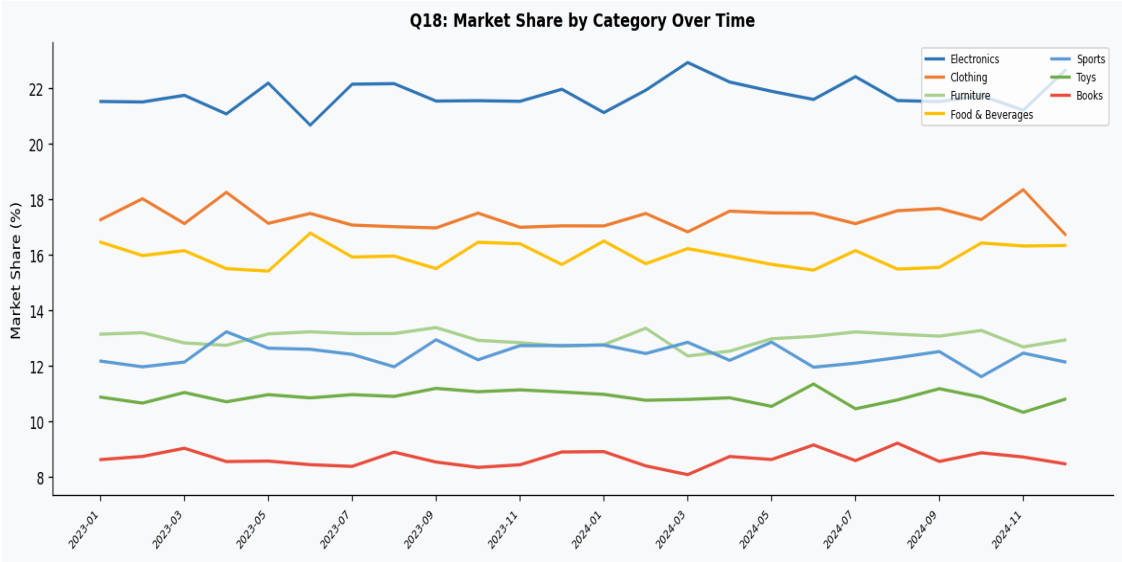
Q18 – Line Chart (Market Share Trend): Market Share Gain and Loss Analysis

Line chart showing each category's market share over 24 months, revealing which categories are gaining or losing ground.

SQL Query

```
WITH MonthlyShare AS (
  SELECT YEAR, MONTH, CATEGORY,
         SUM(SALES)                                          AS CAT_SALES,
         SUM(SUM(SALES)) OVER (PARTITION BY YEAR, MONTH)     AS TOTAL_SALES
  FROM   RETAIL_TRANSACTIONS
  GROUP  BY YEAR, MONTH, CATEGORY
)
SELECT *, ROUND(CAT_SALES*100.0/TOTAL_SALES, 2)             AS MARKET_SHARE_PCT,
       ROUND(CAT_SALES*100.0/TOTAL_SALES -
             LAG(CAT_SALES*100.0/TOTAL_SALES)
               OVER (PARTITION BY CATEGORY
                     ORDER BY YEAR, MONTH), 2)               AS MOM_SHARE_CHANGE
FROM   MonthlyShare
ORDER  BY YEAR, MONTH, CAT_SALES DESC;
```

Visualization: Line Chart (Market Share Trend)



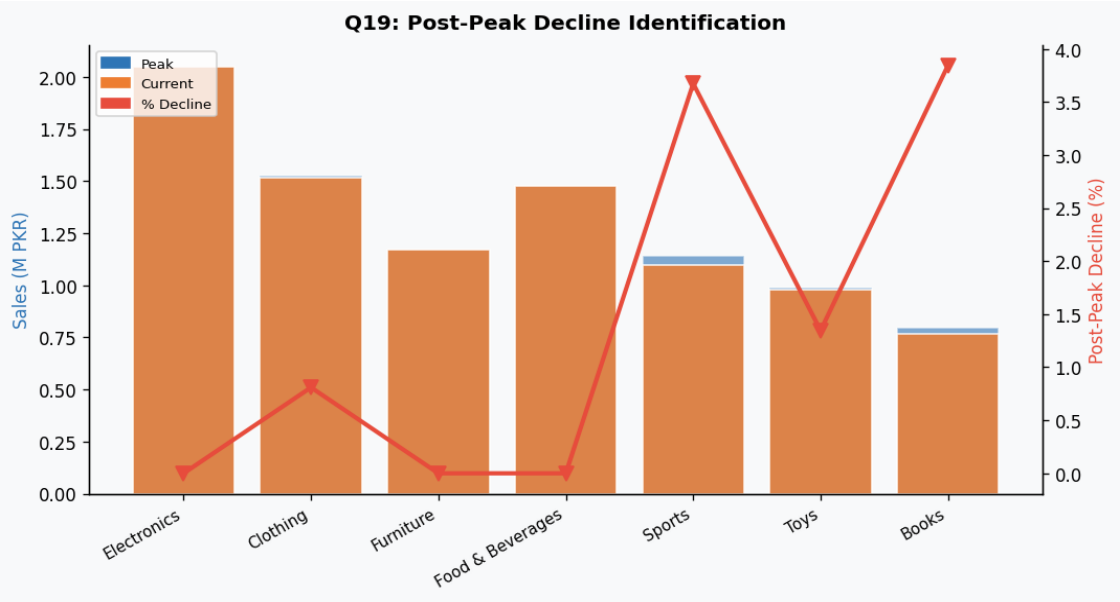
Q19 – Dual-Axis Bar+Line (Post-Peak Decline): Post-Peak Decline Identification

Dual-axis chart showing each category's peak vs current sales as bars, and post-peak decline percentage as a line.

SQL Query

```
WITH PeakSales AS (
  SELECT CATEGORY,
         MAX(MONTHLY_SALES)                                    AS PEAK_SALES
  FROM (
    SELECT CATEGORY, YEAR, MONTH, SUM(SALES) AS MONTHLY_SALES
    FROM   RETAIL_TRANSACTIONS
    GROUP  BY CATEGORY, YEAR, MONTH
  ) t
  GROUP  BY CATEGORY
)
SELECT t.CATEGORY, p.PEAK_SALES,
       SUM(t.SALES)                                          AS RECENT_SALES,
       ROUND(100.0*(p.PEAK_SALES - SUM(t.SALES))
             / p.PEAK_SALES, 2)                              AS PCT_DECLINE
FROM   RETAIL_TRANSACTIONS t
JOIN   PeakSales p ON t.CATEGORY = p.CATEGORY
WHERE  t.YEAR = (SELECT MAX(YEAR) FROM RETAIL_TRANSACTIONS)
GROUP  BY t.CATEGORY, p.PEAK_SALES
ORDER  BY PCT_DECLINE DESC;
```

Visualization: Dual-Axis Bar+Line (Post-Peak Decline)



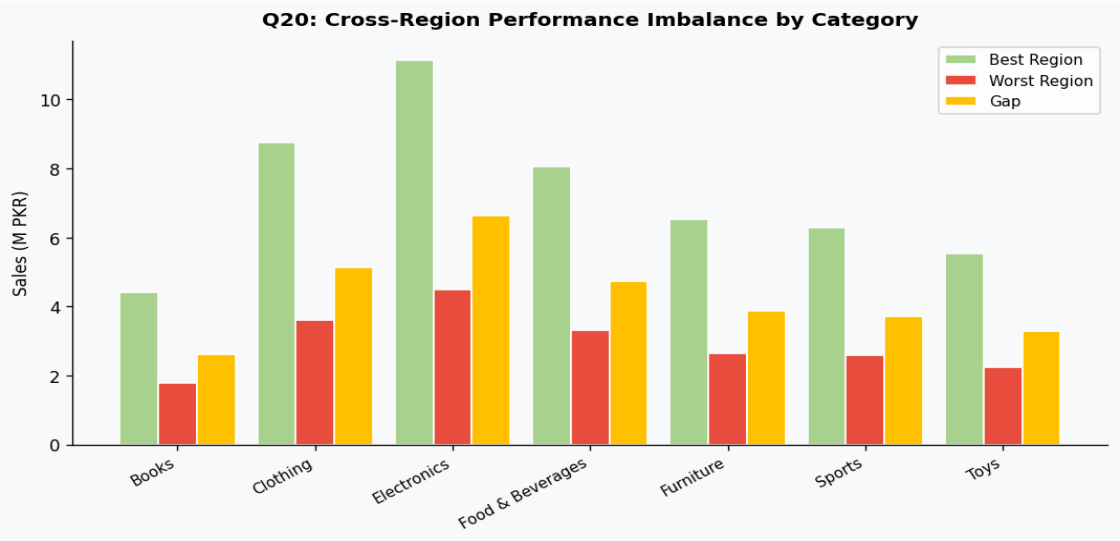
Q20 – Grouped Bar Chart (Regional Imbalance): Cross-Region Performance Imbalance

Grouped bar chart per category showing the best-region sales, worst-region sales, and the resulting performance gap.

SQL Query

```
SELECT CATEGORY,
       MAX(SALES_BY_REGION)                                AS BEST_REGION_SALES,
       MIN(SALES_BY_REGION)                                AS WORST_REGION_SALES,
       MAX(SALES_BY_REGION) - MIN(SALES_BY_REGION)        AS PERFORMANCE_GAP
FROM (
  SELECT CATEGORY, REGION, SUM(SALES) AS SALES_BY_REGION
  FROM   RETAIL_TRANSACTIONS
  GROUP  BY CATEGORY, REGION
) t
GROUP  BY CATEGORY
ORDER  BY PERFORMANCE_GAP DESC;
```

Visualization: Grouped Bar Chart (Regional Imbalance)


Q20: Cross-Region Performance Imbalance by Category
Best Region
Worst Region
Gap
Sales (M PKR)
0
2
4
6
8
10
Books
Clothing
Electronics
Food & Beverages
Furniture
Sports
Toys